4.3 Continuity

In the following let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ and $(Z, \|\cdot\|_Z)$ be normed vector spaces.

Definition 4.3.1 Let $D \subset X$ be nonempty and $f : D \to Y$ a function. f is said to be continuous at $x_0 \in D$ if for every $V \in \mathcal{U}(f(x_0))$ there exists a $U \in \mathcal{U}(x_0)$ with

$$f(U \cap D) \subset V.$$

f is said to be continuous in $D' \subset D$, $D' \neq \varnothing$, if f is continuous at every point $x \in D'$. If f is continuous in D, we simply say f is continuous.

Before we give examples of continuous and non-continuous mappings, we want to state some other equivalent definitions of continuity.

Theorem 4.3.2 Let $f : D \to Y$, $D \subset X$, and $x_0 \in D$. Then the following statements are equivalent:

(1) f is continuous at x_0.

(2) $\forall\, \varepsilon > 0\ \exists\, \delta > 0 :\ x \in D,\ \|x - x_0\|_X < \delta \implies \|f(x) - f(x_0)\|_Y < \varepsilon$.

(3) $\forall\, (x_n)_{n\in\mathbb{N}} \subset D$ with $\lim_{n\to\infty} x_n = x_0$ it holds:

$$\lim_{n\to\infty} f(x_n) = f(x_0).$$

Proof.

a) We show $1 \implies 2$:
Let $\varepsilon > 0$ and set $V := B(f(x_0), \varepsilon) \in \mathcal{U}(f(x_0))$. Then by Def. 4.2.1 there exists $U \in \mathcal{U}(x_0)$ with

$$f(U \cap D) \subset V.$$

Since U is open, by Def. 4.2.1 there is a $\delta > 0$ with

$$B(x_0, \delta) \subset U.$$

Hence

$$f\big(B(x_0, \delta) \cap D\big) \subset f(U \cap D) \subset V$$

exists for all $\varepsilon > 0$, i.e. there is $\delta > 0$, so that for all $x \in D$ with $\|x - x_0\|_X < \delta$ it holds:

$$\|f(x) - f(x_0)\|_Y < \varepsilon.$$

b) We show $2 \implies 3$:
Let $(x_n)_{n\in\mathbb{N}} \subset D$ with $x_n \xrightarrow{n\to\infty} x_0$ and let $\varepsilon > 0$. Let δ be as in (2). Then there is an $n_\varepsilon \in \mathbb{N}$ with

$$\|x_n - x_0\|_X < \delta \qquad \forall n \geq n_\varepsilon.$$

Therefore

$$f(x_0) = \lim_{n\to\infty} f(x_n).$$

c) We show 3 $\implies$ 1:
Assume (3). Suppose f is *not* continuous at x_0. Then (negation of the definition) it follows:

$$\begin{aligned}
&\exists V \in \mathcal{U}(f(x_0))\ \forall U \in \mathcal{U}(x_0): \quad f(U \cap D) \nsubseteq V \\
\implies &\exists V \in \mathcal{U}(f(x_0))\ \forall n \in \mathbb{N}: \ f\big(B(x_0, \tfrac{1}{n}) \cap D\big) \nsubseteq V \\
\implies &\exists V \in \mathcal{U}(f(x_0))\ \forall n \in \mathbb{N}: \ \exists x_n \in B(x_0, \tfrac{1}{n}) \cap D: \ f(x_n) \notin V \\
\implies &\exists (x_n)_{n\in\mathbb{N}} \subset D, \ \lim_{n\to\infty} x_n = x_0 \text{ and } f(x_n) \nrightarrow f(x_0).
\end{aligned}$$

Contradiction to (3). Thus (1) follows.

■

Theorem 4.3.3 Let $f : D \to Y$ with $D \subset X$ and $\varnothing \neq D' \subset D$. Then the following statements are equivalent:

(1) f is continuous on D'.

(2) Preimages of open sets in Y are relatively open in D', i.e. for every open subset $V \subset Y$ the preimage

$$f^{-1}(V) := \{x \in D : f(x) \in V\}$$

has the property that $D' \cap f^{-1}(V)$ is relatively open in D'.

(3) Preimages of closed sets in Y are relatively closed in D', i.e. for every closed subset $A \subset Y$ it holds: $D' \cap f^{-1}(A)$ is relatively closed in D'.

Proof.

a) We show 1 $\implies$ 2:
Consider[1] $D' \cap f^{-1}(V)$.
extbf1st case: $\varnothing = D' \cap f^{-1}(V)$ is relatively open.
extbf2nd case: Let $x \in D' \cap f^{-1}(V)$. We claim: There exists an open set $U \in \mathcal{U}(x)$ with $D' \cap U \subset f^{-1}(V) \cap D'$.
From $x \in f^{-1}(V)$ it follows

$$f(x) \in V \implies V \in \mathcal{U}(f(x)) \implies \exists U \in \mathcal{U}(x): \ f(U \cap D') \subset V.$$

Hence

$$U \cap D' \subset f^{-1}(V).$$

This shows $x \in U \cap D'$, and x is a relatively interior point of $f^{-1}(V) \cap D'$.

b) We show 2 $\implies$ 3:
Let $A \subset Y$ be closed. Then $Y \setminus A$ is open and, by (2),

$$D' \setminus f^{-1}(A) = D' \cap f^{-1}(Y \setminus A)$$

is relatively open in D', i.e. there exists an open subset U in X with

$$D' \setminus f^{-1}(A) = U \cap D' \quad \implies \quad f^{-1}(A) \cap D' = (X \setminus U) \cap D'.$$

Since $X \setminus U$ is closed in X, it follows that $f^{-1}(A) \cap D'$ is relatively closed in D'.

[1] Reminder: In the special case $D' = X$, "relatively open" means the same as "open", i.e., $f^{-1}(V)$ is open exactly when for each $x \in f^{-1}(V)$ there exists a $U \in \mathcal{U}(x)$ with $U \subset f^{-1}(V)$. In the general case $\varnothing \neq D' \subset D \subset X$, the set $f^{-1}(V) \cap D'$ is relatively open in D' exactly when for every $x \in f^{-1}(V) \cap D'$ there exists a $U \in \mathcal{U}(x)$ with $U \cap D' \subset f^{-1}(V) \cap D'$.

c) We show 3 $\implies$ 1:
Let $x \in D'$ and $V \in \mathcal{U}(f(x))$. Then $Y \setminus V$ is closed and, by (3),

$$D' \setminus f^{-1}(V) = D' \cap f^{-1}(Y \setminus V)$$

is relatively closed in D'. As in the proof of (2) $\implies$ (3) it follows that $f^{-1}(V)$ is relatively open in D'. Hence there exists an open set U with

$$x \in U \quad \text{and} \quad f^{-1}(V) \cap D' = U \cap D'.$$

Therefore

$$f(U \cap D') = f\big(f^{-1}(V) \cap D'\big) \subset f\big(f^{-1}(V)\big) = V.$$

Thus f is continuous.

■

Example 4.3.4

(1) $f : X \to \mathbb{R}_{\geq 0}$ with $f(x) := \|x\|$ is continuous.
(2) $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $f(x) := \sqrt{x}$ is continuous.
(3) $f : \mathbb{R} \to \mathbb{R}$ with $f(x) := \lfloor x \rfloor := \max\{k \in \mathbb{Z} : k \leq x\}$ is continuous for $x \in \mathbb{R} \setminus \mathbb{Z}$ and not continuous for $x \in \mathbb{Z}$.
(4) $f : \mathbb{R} \to \mathbb{R}$ with

$$f(x) := \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}, \end{cases}$$

is continuous at no point $x \in \mathbb{R}$.
(5) $f : \mathbb{K}^m \to \mathbb{K}$ with $f(x) := x_j$ for a fixed $j \in \{1, 2, \ldots, m\}$ is continuous for all $x \in \mathbb{K}^m$.
(6) The function $f : \mathbb{C} \to \mathbb{R}$ with $f(z) := \mathrm{Re}(z)$ is continuous. The same holds for $f(z) := \mathrm{Im}(z)$.
(7) Let $(X, \|\cdot\|_a)$, $(X, \|\cdot\|_b)$ be two normed vector spaces with equivalent norms. Then the identity $I : (X, \|\cdot\|_a) \to (X, \|\cdot\|_b)$, $Ix := x$, is continuous.
(8) Let $f : X \to \mathbb{R}$ be continuous and $r \in \mathbb{R}$. Then the sets

$$\{x \in X : f(x) > r\},\ \{x \in X : f(x) < r\} \text{ are open and}$$

$$\{x \in X : f(x) \geq r\},\ \{x \in X : f(x) \leq r\} \text{ are closed.}$$

Proof.

(1) For $\varepsilon > 0$ choose $\delta = \varepsilon$. Then for all $x \in X$ with $\|x - x_0\| < \delta$ we have

$$\big|\, \|x_0\| - \|x\| \,\big| \leq \|x - x_0\| < \varepsilon.$$

(2) a) $x_0 = 0$: Let $\varepsilon > 0$ and $\delta := \varepsilon^2$. Then for all $x \in [0, \delta]$ we have

$$\big|\sqrt{x} - \sqrt{x_0}\big| = \sqrt{x} < \sqrt{\delta} = \varepsilon.$$

b) $x_0 > 0$: Let $\varepsilon > 0$ and $\delta := \varepsilon\sqrt{x_0}$. Then

$$\big|\sqrt{x} - \sqrt{x_0}\big| = \left|\frac{x - x_0}{\sqrt{x} + \sqrt{x_0}}\right| \leq \frac{1}{\sqrt{x_0}}|x - x_0| < \frac{\varepsilon\sqrt{x_0}}{\sqrt{x_0}} = \varepsilon.$$

(3) If $x_0 \notin \mathbb{Z}$, then there exists $\delta > 0$ with $I = (x_0 - \delta, x_0 + \delta) \subset \mathbb{R} \setminus \mathbb{Z}$. On I the function $f(x) = \lfloor x \rfloor$ is constant and hence continuous.
If $x_0 \in \mathbb{Z}$, then

$$\forall x < x_0 : \; f(x) \leq f(x_0) - 1 = x_0 - 1, \qquad \forall x \geq x_0 : \; f(x) \geq f(x_0) = x_0.$$

Therefore for every $\delta > 0$ there exists an $x \in (x_0 - \delta, x_0 + \delta)$ with

$$|f(x) - f(x_0)| \geq 1.$$

(4) For each $x \in \mathbb{R} \setminus \mathbb{Q}$ there exists a sequence $(x_n) \subset \mathbb{Q}$ with $x_n \to x$. Then

$$f(x) = 0, \qquad \lim_{n \to \infty} f(x_n) = 1.$$

For $x \in \mathbb{Q}$ we can construct a sequence $(x_n) \subset \mathbb{R} \setminus \mathbb{Q}$ with $x_n \to x$ and argue as before.

(5) Since all norms on $\mathbb{K}^m$ are equivalent, it holds:

$$|f(x_0) - f(x)| = |(x_0)_j - x_j| \leq \max_{1 \leq i \leq m} |(x_0)_i - x_i| = \|x_0 - x\|_\infty \leq C\|x_0 - x\|.$$

For $\varepsilon > 0$ choose $\delta = \frac{\varepsilon}{C}$.

(6) Continuity follows from $|\operatorname{Re}(z_0) - \operatorname{Re}(z)| = |\operatorname{Re}(z_0 - z)| \leq |z_0 - z|$ with $\delta = \varepsilon$.

(7) We have:

$$c\|x\|_a \leq \|x\|_b \leq c'\|x\|_a \qquad \forall x \in X.$$

Let $\varepsilon > 0$ and $\delta := \frac{\varepsilon}{c'}$. Then

$$\|Ix_0 - Ix\|_b = \|x_0 - x\|_b \leq c'\|x_0 - x\|_a < \varepsilon.$$

(8) We have

$$B := \{y \in \mathbb{R} : y < r\} \text{ is open and } \{x \in X : f(x) < r\} = f^{-1}(B).$$

Therefore $f^{-1}(B)$ is open (see Theorem 4.3.3). The remaining cases in (8) are treated analogously.

■

Theorem 4.3.5 Let $D_f, D_g \subset X$, $x_0 \in D_f \cap D_g$ and $f : D_f \to Y$, $g : D_g \to Y$ be continuous at x_0, as well as $\alpha \in \mathbb{K}$. Then:

(1) $\alpha f : x \mapsto \alpha \cdot f(x)$ is continuous at x_0.
(2) $f + g : x \mapsto f(x) + g(x)$ is continuous at x_0.
(3) $Y = \mathbb{K}$, $f \cdot g : x \mapsto f(x) \cdot g(x)$ is continuous at x_0.
(4) $Y = \mathbb{K}$ and $g(x_0) \neq 0$. Then $\frac{f}{g} : x \mapsto \frac{f(x)}{g(x)}$ is continuous at x_0.

Proof. Use the sequence criterion (Theorem 4.3.2) for characterization of continuity to reduce the statements back to Theorem 3.1.19 (algebraic rules with sequences). ■

Definition 4.3.6 Let $M \subset X$, $M \neq \varnothing$. Then $C(M, Y)$ denotes the vector space of all continuous functions from M to Y.

Theorem 4.3.5 and Definition 4.3.6 imply that the set of all continuous functions $f : M \to Y$, i.e. $C(M, Y)$, forms a vector space.

Theorem 4.3.7 All polynomials, i.e. functions of the form

$$f(x) = \sum_{k=0}^{n} a_k x^k, \qquad a_k \in \mathbb{K},\ n \in \mathbb{N},$$

and all rational functions, i.e. functions of the form

$$f(x) = \frac{p(x)}{q(x)}, \qquad p, q \text{ polynomials},$$

are continuous on their domain of definition.

Proof. Immediate consequence of Theorem 4.3.5. ■

Theorem 4.3.8 Let $f : D_f \to Y$, $D_f \subset X$ and $g : D_g \to Z$, $D_g \subset Y$ be two functions such that $f(D_f) \subset D_g$. Then the composition $g \circ f : D_f \to Z$ defined by

$$(g \circ f)(x) := g(f(x))$$

is continuous at $x_0 \in D_f$ if f is continuous at x_0 and g is continuous at $f(x_0)$.

Proof. Let $W \in \mathcal{U}(g(y_0))$. Since g is continuous at y_0, there exists $V \in \mathcal{U}(y_0)$ with

$$g(V \cap D_g) \subset W.$$

Since f is continuous at x_0, there exists $U \in \mathcal{U}(x_0)$ with

$$f(U \cap D_f) \subset V \quad \text{and hence} \quad f(U \cap D_f) \subset V \cap D_g.$$

It follows:

$$(g \circ f)(U \cap D_f) \subset W.$$

■

Example 4.3.9

(1) Let $f : D \to Y$, $D \subset X$ be continuous. Then $\|f\|_Y : D \to \mathbb{R}_{\geq 0}$ with $x \mapsto \|f(x)\|_Y$ is continuous (see Example 4.3.4(1), Theorem 4.3.8).

(2) Let $f : D \to \mathbb{K}$, $D \subset X$ be continuous. Then $f^2 : D \to \mathbb{K}$ with $x \mapsto (f(x))^2$ is continuous (Theorem 4.3.5(3)).

(3) Let $f : D \to \mathbb{R}_{\geq 0}$, $D \subset X$ be continuous. Then $\sqrt{f} : D \to \mathbb{R}_{\geq 0}$ with $x \mapsto \sqrt{f(x)}$ is continuous (see Example 4.3.4(2), Theorem 4.3.8).

Theorem 4.3.10

(1) $f = (f_1, f_2, \ldots, f_n) : D \to \mathbb{K}^n$, $D \subset X$, is continuous at $x_0 \in D$ if and only if all component functions f_i, $1 \leq i \leq n$, are continuous at x_0.

(2) $f : D \to \mathbb{C}$, $D \subset X$, is continuous at $x_0 \in D$ if and only if $\operatorname{Re} f$ and $\operatorname{Im} f$ are continuous at x_0.

Proof.

(1) The direction "$\Rightarrow$" follows from Theorem 4.3.8 and Example 4.3.4(5) (the map $(x_1, \dots, x_n) \mapsto x_i$ is continuous).
For the converse direction, let $\varepsilon > 0$. By the assumption there exist numbers $\delta_j(\varepsilon)$, $1 \le j \le n$, with

$$|f_j(x) - f_j(x_0)| < \frac{\varepsilon}{n} \qquad \text{whenever } \|x - x_0\|_X < \delta_j(\varepsilon).$$

Set $\delta := \min_{1 \le j \le n} \delta_j(\varepsilon)$. Then for $x \in B(x_0, \delta)$ we obtain

$$\|f(x) - f(x_0)\|_1 = \sum_{j=1}^{n} |f_j(x) - f_j(x_0)| < \varepsilon.$$

To prove continuity with respect to any other norm on $\mathbb{K}^n$, use the equivalence of norms and replace ε/n by $\varepsilon/(nc)$ with the corresponding equivalence constant c.

(2) The claim follows from

$$|f(z)| = \sqrt{(\operatorname{Re} f(z))^2 + (\operatorname{Im} f(z))^2} = \left\| \begin{pmatrix} \operatorname{Re} f(z) \\ \operatorname{Im} f(z) \end{pmatrix} \right\|_2,$$

together with part (1) and Example 4.3.4(6).

■

Definition 4.3.11 Let $f : D \to Y$, $D \subset X$, be a function. Let $x_0 \in X$ and suppose there exists a sequence $(x_n) \subset D$ with $\lim_{n\to\infty} x_n = x_0$. We define

$$y := \lim_{x \to x_0} f(x),$$

if for every sequence $(x_n) \subset D$ with $x_n \to x_0$ as $n \to \infty$ it holds that

$$f(x_n) \longrightarrow y.$$

Theorem 4.3.12 The following statements are equivalent:
(1) $y = \lim_{x\to x_0} f(x)$,
(2) $\forall V \in \mathcal{U}(y) : \ \exists U \in \mathcal{U}(x_0) : \ f(D \cap U) \subset V$.

Proof.

a) We show (1) $\Rightarrow$ (2) indirectly and assume instead that

$$\exists V \in \mathcal{U}(y) \ \forall U \in \mathcal{U}(x_0) : \ f(D \cap U) \not\subseteq V.$$

Then it follows that for all $n \in \mathbb{N}$,

$$f\Big(D \cap B\Big(x_0, \tfrac{1}{n}\Big)\Big) \not\subseteq V,$$

and hence there exist $x_n \in D \cap B\big(x_0, \frac{1}{n}\big)$ with $f(x_n) \notin V$. That is, there exists a sequence $(x_n)_{n\in\mathbb{N}} \subset D$ converging to x_0, but with $f(x_n) \notin V$.

b) We show (2) $\Rightarrow$ (1):
Let $(x_n)_{n\in\mathbb{N}} \subset D$ with $x_n \to x_0$ and let $\varepsilon > 0$. Then there exists a $U \in \mathcal{U}(x_0)$ with $f(D \cap U) \subset B(y, \varepsilon)$. Since $x_n \to x_0$ there exists an $n_U \in \mathbb{N}$ with

$$x_n \in U \qquad \forall n \geq n_U.$$

Hence

$$f(x_n) \in B(y, \varepsilon) \qquad \forall n \geq n_U,$$

and therefore $f(x_n) \to y$.

■

Remark 4.3.13

(1) Let $f : D \to Y$, $D \subset X$, and $x_0 \in D$. Then: "$\lim_{x\to x_0} f(x) = f(x_0)$" is equivalent to "$f$ is continuous at x_0". (Theorem 4.3.2)

(2) Let $f : D \to Y$, $D \subset X$, $x_0 \notin D$ and suppose

$$y = \lim_{x\to x_0} f(x).$$

Then we define an extension of f by $\tilde{f} : D \cup \{x_0\} \to Y$ through

$$\tilde{f}(x) := \begin{cases} f(x), & x \in D, \\ y, & x = x_0. \end{cases}$$

The function $\tilde{f}$ is continuous at x_0 and is therefore called a continuous extension.

Example 4.3.14

(1) $X = Y = \mathbb{R}$, $D = X \setminus \{1\}$ and $f(x) := \frac{x^n-1}{x-1}$, $n \in \mathbb{N}$. From Lemma 3.2.3 it follows

$$f(x) = \sum_{k=0}^{n-1} x^k \qquad \forall x \in \mathbb{R} \setminus \{1\},$$

and hence

$$\lim_{x\to 1} f(x) = n.$$

That is,

$$\tilde{f}(x) = \begin{cases} \frac{x^n-1}{x-1}, & x \in \mathbb{R} \setminus \{1\}, \\ n, & x = 1 \end{cases}$$

is a continuous extension of f.

(2) $X, Y = \mathbb{C}$, $D \subset \mathbb{C} \setminus \{0\}$ and $f(z) = \frac{e^z-1}{z}$. We show: For any sequence (z_n) with $z_n \to 0$, $f(z_n) \to 1$. This follows from:

$$|f(z) - 1| = \left| \sum_{n=1}^{\infty} \frac{z^{n-1}}{n!} - 1 \right| = \left| \sum_{n=2}^{\infty} \frac{z^{n-1}}{n!} \right| = \left| \sum_{m=1}^{\infty} \frac{z^m}{(m+1)!} \right| \leq \frac{1}{2} \sum_{m=1}^{\infty} |z|^m$$

$$= \frac{1}{2}\Big(\frac{1}{1-|z|} - 1\Big) = \frac{|z|}{2(1-|z|)} \longrightarrow 0 \qquad |z| < 1.$$

■

Definition 4.3.15 The function $f : D \to \mathbb{R}$, $D \subset \mathbb{R}$ is called left-continuous (resp. right-continuous) at $x_0 \in D$ if for every $V \in \mathcal{U}(f(x_0))$ there exists a $\delta > 0$ with

$$f(D \cap]x_0 - \delta, x_0]) \subset V \qquad (\text{resp. } f(D \cap [x_0, x_0 + \delta[) \subset V).$$

Example 4.3.16 The function $\lfloor \cdot \rfloor$ from Example 4.3.4(3) is right-continuous at every $x \in \mathbb{R}$.

Theorem 4.3.17 The function $f : D \to \mathbb{R}$, $D \subset \mathbb{R}$, is continuous at $x_0 \in D$ if and only if it is both right- and left-continuous at x_0.

Proof. "$\Rightarrow$": Let f be continuous at x_0. Then there exists for all $V \in \mathcal{U}(f(x_0))$ a $U \in \mathcal{U}(x_0)$ with $f(D \cap U) \subset V$. Then there exists a $\delta > 0$ with $]x_0 - \delta, x_0] \subset U$. It follows

$$f(D \cap]x_0 - \delta, x_0]) \subset f(D \cap U) \subset V.$$

The analogous argument gives right-continuity.

"$\Leftarrow$": Let f be left- and right-continuous at x_0, i.e. there exist $\delta_1, \delta_2 > 0$ with

$$f(D \cap]x_0 - \delta_1, x_0]) \subset V, \qquad f(D \cap [x_0, x_0 + \delta_2[) \subset V.$$

Choose $\delta := \min\{\delta_1, \delta_2\} > 0$. Then

$$f\big(D \cap]x_0 - \delta, x_0 + \delta[\big) \subset V.$$

■

In many cases it is important when, for prescribed $\varepsilon > 0$, for each point $x \in D$, the $\delta(x, \varepsilon)$ in the definition does not depend on x.

Definition 4.3.18 Let $f : D \to Y$, $D \subset X$, be a function and $D' \subset D$. Then f is called uniformly continuous on D' if for every $\varepsilon > 0$ there exists a $\delta > 0$ with

$$\|f(x) - f(y)\|_Y < \varepsilon \qquad \forall x, y \in D' \text{ with } \|x - y\|_X < \delta.$$

Remark 4.3.19 A uniformly continuous function is continuous.

Example 4.3.20

(1) $f : X \to \mathbb{R}_{\geq 0}$ with $f(x) := \|x\|$ is uniformly continuous.
(2) $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ with $f(x) := \frac{1}{x}$ is not uniformly continuous.
(3) $f : [-1, 1] \to \mathbb{R}$ with $f(x) := x^2$ is uniformly continuous.
(4) $f : \mathbb{R} \to \mathbb{R}$ with $f(x) := x^2$ is not uniformly continuous.

Proof.

1. Use the reverse triangle inequality:

$$\big|\, \|x\| - \|y\| \,\big| \le \|x - y\|$$

for all x, y, hence $x \mapsto \|x\|$ is uniformly continuous.

2. By Theorem 4.3.5 the function $f(x) = \frac{1}{x}$ is continuous. Fix $\varepsilon := 1$ and let $\delta > 0$ be arbitrary. Choose

$$x = \frac{1}{n}, \qquad n > \frac{1}{2\delta},\ n \in \mathbb{N}, \qquad \text{and} \qquad y = \frac{1}{2n}.$$

Then $|x - y| = \frac{1}{2n} < \delta$, but

$$|f(x) - f(y)| = \left|\frac{1}{x} - \frac{1}{y}\right| = |n - 2n| = n > 1 = \varepsilon.$$

Hence f is not uniformly continuous on $\mathbb{R} \setminus \{0\}$.

3. For $x, y \in [-1, 1]$,

$$|x^2 - y^2| = |x - y| \cdot |x + y| \le 2\,|x - y|.$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon/2$. Then $|x - y| < \delta$ implies $|x^2 - y^2| < \varepsilon$, so $x \mapsto x^2$ is uniformly continuous on $[-1, 1]$.

4. The function $x \mapsto x^2$ is continuous since it is a polynomial. To see it is not uniformly continuous on $\mathbb{R}$, fix $\varepsilon := 1$ and let $\delta > 0$ be arbitrary. Choose $x = n$ with $n > \frac{1}{\delta}$, $n \in \mathbb{N}$, and set $y = x + \frac{\delta}{2}$. Then

$$|x - y| = \frac{\delta}{2} < \delta, \qquad |x^2 - y^2| = \left|x\delta + \tfrac{\delta^2}{4}\right| \ge |x\delta| = n\delta > 1 = \varepsilon.$$

Therefore $x \mapsto x^2$ is not uniformly continuous on $\mathbb{R}$.

$\square$